

Winter 2026 DSC 240: Homework 3

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Assignment 1 (10 points): Maximum Likelihood Estimation for Bernoulli Distribution

Let $X_1, X_2, \dots, X_n$ be n independent and identically distributed (i.i.d.) random variables following a Bernoulli distribution. The Bernoulli distribution for a random variable X is given by:

$$\mathbb{P}(X = x; p) = p^x(1 - p)^{1-x}$$

where $x \in \{0, 1\}$ and p is the probability of X being 1. Derive the maximum likelihood estimator (MLE) for the parameter p using the given samples.

Proof. We directly calculate the log-likelihood:

$$\ell(p) = \log L(p) = \sum_{i=1}^n [x_i \log p + (1 - x_i) \log(1 - p)].$$

Take the derivative w.r.t to p and let it to be zero:

$$\frac{d\ell}{dp} = \sum_{i=1}^n \left(\frac{x_i}{p} - \frac{1 - x_i}{1 - p} \right) = 0.$$

We get the MLE estimator

$$\hat{p} = \frac{\sum_{i=1}^n x_i}{n}.$$

Assignment 2 (20 points): *Cross-Entropy Loss and Logistic Loss* Consider a binary classification problem with outputs $y \in \{0, 1\}$. Assume $p(y \mid x)$ is Bernoulli ($\hat{p}(x; \theta^*)$) for some parameter vector $\theta^* \in \mathbb{R}^d$, where for each θ , $\hat{p}(x; \theta)$ returns a number between $[0, 1]$.

(a) (10 points) For each parameter θ , the cross-entropy loss for a set of n examples with predictions $\hat{p}_i := \hat{p}(x_i; \theta)$ and true labels y_i is given by:

$$\text{CE}(\theta) = -\frac{1}{n} \sum_{i=1}^n (y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)).$$

Derive this loss function starting from the principle of Maximum Likelihood Estimation (MLE).

(b) (10 points) Instantiate the classifier's prediction $\hat{p}(x; \theta)$ as the sigmoid function applied to a linear combination of input features, i.e., $\hat{p}(x; \theta) = \sigma(\theta^\top x)$, where $\sigma(z) = \frac{1}{1+e^{-z}}$. Show that in this case, minimizing the cross-entropy loss above is equivalent to minimizing the familiar logistic loss, which is given by:

$$L(\theta) = \frac{1}{n} \sum_{i=1}^n \log \left(1 + e^{-\tilde{y}_i \theta^\top x_i} \right) \quad \text{where } \tilde{y}_i = \begin{cases} 1 & \text{if } y_i = 1 \\ -1 & \text{if } y_i = 0 \end{cases}.$$

Proof. (a) The likelihood over the dataset is

$$L(\theta) = \prod_{i=1}^n \hat{p}_i^{y_i} (1 - \hat{p}_i)^{1-y_i}.$$

Maximize log-likelihood $\Leftrightarrow$ minimize negative average log-likelihood:

$$-\frac{1}{n} \log L(\theta) = -\frac{1}{n} \sum_{i=1}^n (y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)),$$

which is exactly the cross-entropy expression.

(b) Plug in the expression of $\hat{p}$ and unify the case for y_i , we have:

$$\text{CE}(\theta) = \frac{1}{n} \sum_{i=1}^n (\log(1 + e^{-\theta^\top x_i}) I_{y_i=1} + \log(1 + e^{\theta^\top x_i}) I_{y_i=0}) \quad (0.1)$$

$$= \frac{1}{n} \sum_{i=1}^n \log \left(1 + e^{-\tilde{y}_i \theta^\top x_i} \right) \quad (0.2)$$

$$= L(\theta). \quad (0.3)$$

Assignment 3 (40 + 10 bonus points): Shape of Bayes Optimal Classifiers for Gaussian Data

Assume we have a binary classification problem where:

- Data points from class A are generated from a Gaussian distribution $\mathcal{N}(\mu_1, I_d)$.
- Data points from class B are generated from a Gaussian distribution $\mathcal{N}(\mu_2, I_d)$.
- The prior probabilities for class A and class B are equal, $P(\text{class A}) = P(\text{class B}) = 0.5$.

(a) (10 points) Derive the posterior probability $P(y | x)$ for class A and B and the decision rule for the Bayes optimal classifier. Given $\mu_1 = [0, 0]$ and $\mu_2 = [1, 1]$, plot the decision boundary and determine if it is linear. (Hint: Recall that the Bayes optimal classifier h^* is given by $h^*(x) = \arg \max_y p(y | x)$. So the strategies for solving Part (a) and (b)(c)(d) is to first derive $p(y | x)$ using the Bayes rule.)

(b) (10 points) Consider the case where the prior probabilities are not equal, $P(\text{class A}) = p$ and $P(\text{class B}) = 1 - p$. Derive the Bayes optimal classifier. For the 2D example, let $\mu_1 = [0, 0]^T$, $\mu_2 = [1, 1]^T$, and $p = 0.8$. Plot the decision boundary.

(c) (10 points) Now let the covariance matrix for class A be $\sigma^2 I_d$ while keeping class B's covariance matrix as I_d . Derive the Bayes optimal classifier. For the 2D example, let $\mu_1 = [0, 0]^T$, $\mu_2 = [1, 1]^T$, $p = 0.5$, and $\sigma^2 = 0.25$. Plot the decision boundary.

(d) (10 points) Instantiate the decision rule from (c) with $\mu_1 = \mu_2$ what is the shape of the decision boundary? For the scenario where $\mu_1 = [0, 0]^T$ and $\mu_2 = [0, 0]^T$ with $p = 0.5$ and $\sigma^2 = 0.25$, plot the decision boundary.

(e) (Bonus 10 points) Examine whether the decision boundaries from parts (a)-(d) can be realized by a Gaussian Naive Bayes model. Provide a reasoning for your answer. (Hint: Whether the Bayes Optimal classifier can be realized by $\mathcal{H}$ means whether $h^* \in \mathcal{H}$, i.e., whether there is a parameter vector of the Gaussian Naive Bayes classifier that is the same as h^*)

Deliverables for each subproblem (a) (b) (c) and (d):

1. Analytical expressions for $P(y | x)$.
2. The decision rule of the Bayes optimal classifier / analytical expression of the decision boundary.
3. Plotted figures that clearly demonstrate the decision boundary along with 100 random samples (color-coded in red and blue) from each class in the same figure generated in the specified way.

Solution. Bayes rule:

$$P(A | x) = \frac{\pi_1 p(x | A)}{\pi_1 p(x | A) + \pi_2 p(x | B)}, \quad P(B | x) = 1 - P(A | x).$$

and the optimal solution

$$h^*(x) = \arg \max_{y \in \{A, B\}} P(y | x).$$

(a) For Gaussian, we have

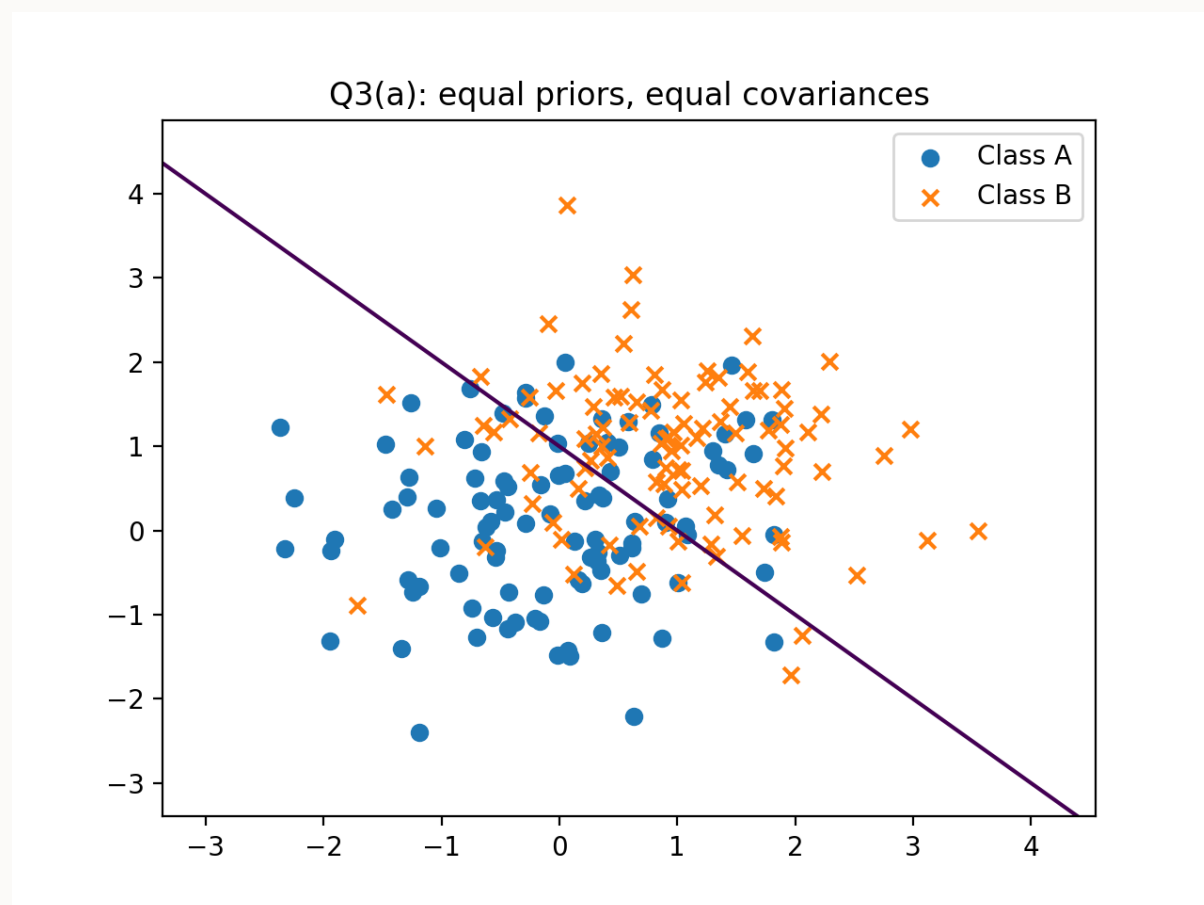
$$\log p(x | y) = -\frac{d}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma_y| - \frac{1}{2} (x - \mu_y)^\top \Sigma_y^{-1} (x - \mu_y).$$

Therefore, the equivalent discriminant rule:

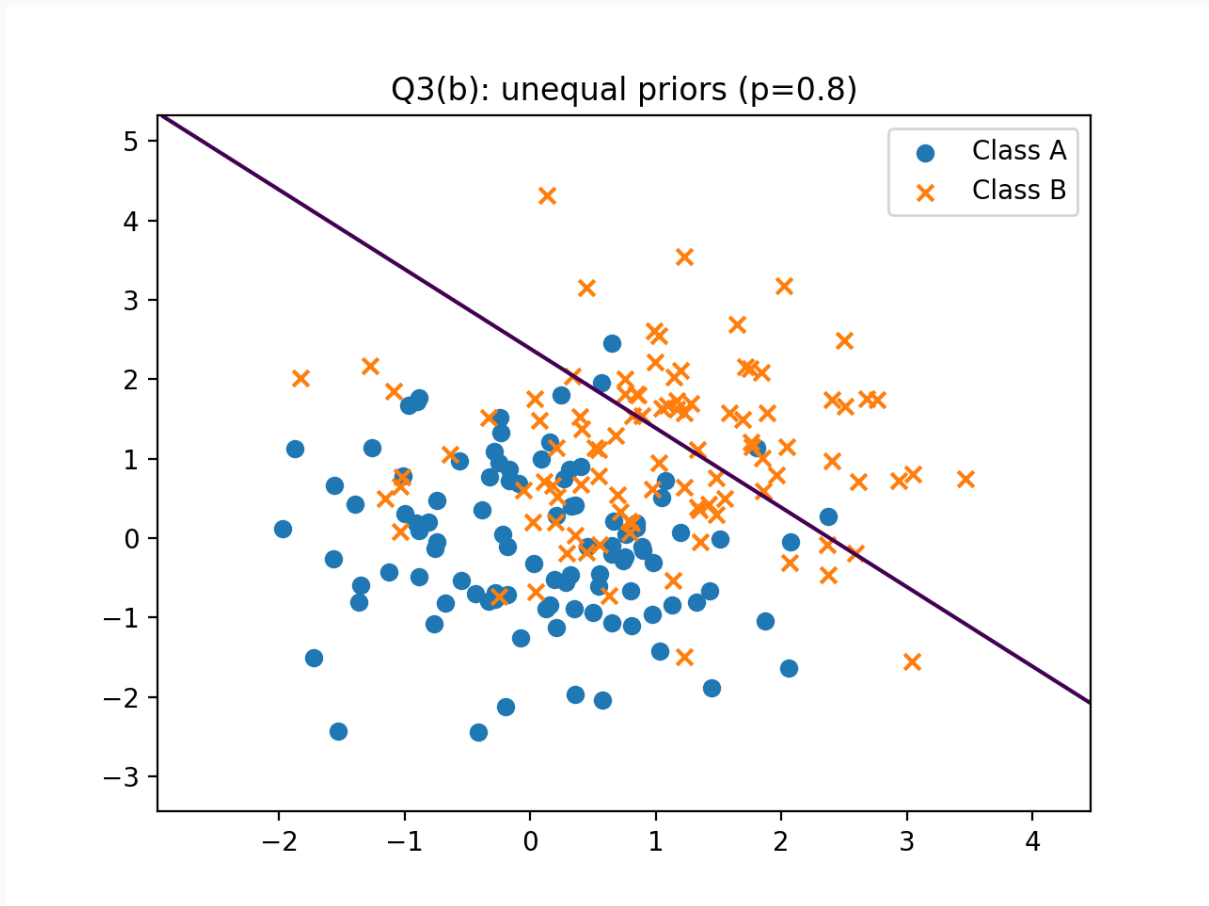
Choose A if $\log \pi_1 + \log p(x | A) > \log \pi_2 + \log p(x | B)$.

Since $\pi_1 = \pi_2$, $\Sigma_A = \Sigma_B$, we see the quadratic terms and prior terms cancel out. Therefore the decision boundary is linear.

$$p(A|x) = \frac{1}{1 + e^{x^\top (\mu_B - \mu_A) + \frac{1}{2}(\|\mu_A\|^2 - \|\mu_B\|^2)}}, p(B|x) \text{ is obvious by symmetry.}$$



(b) Decide A if $\delta(x) > 0$, else decide B, where $\delta(x) = (\mu_1 - \mu_2)^\top x - \frac{1}{2} (\|\mu_1\|^2 - \|\mu_2\|^2) + \log \frac{p}{1-p}$.

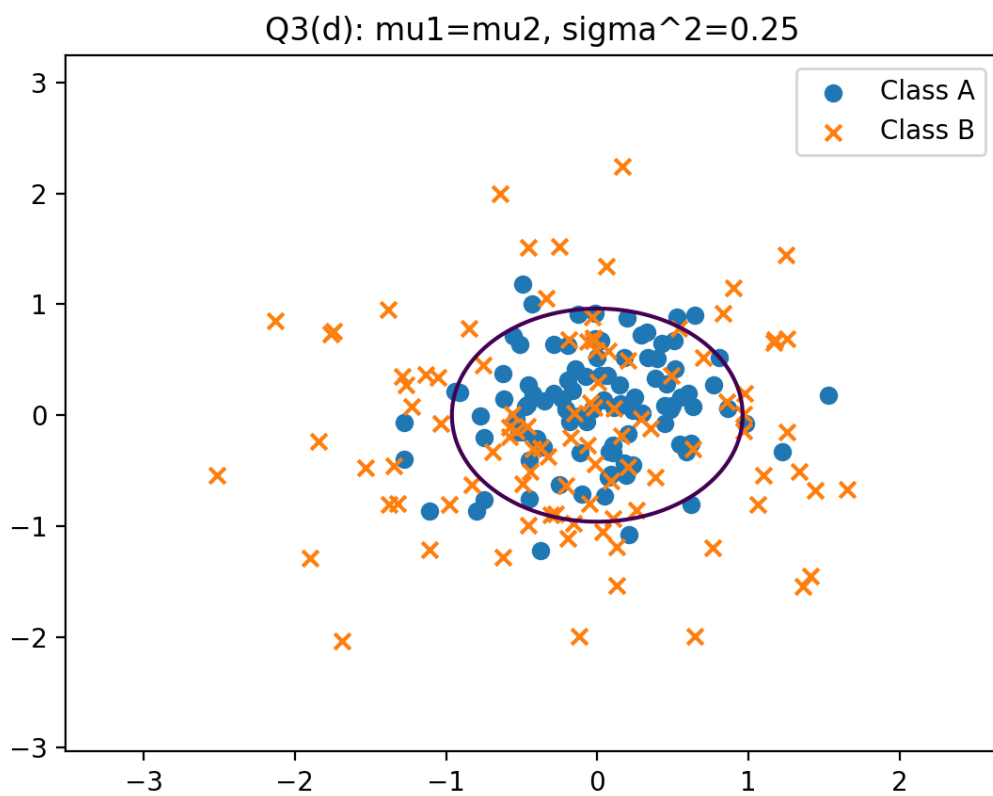
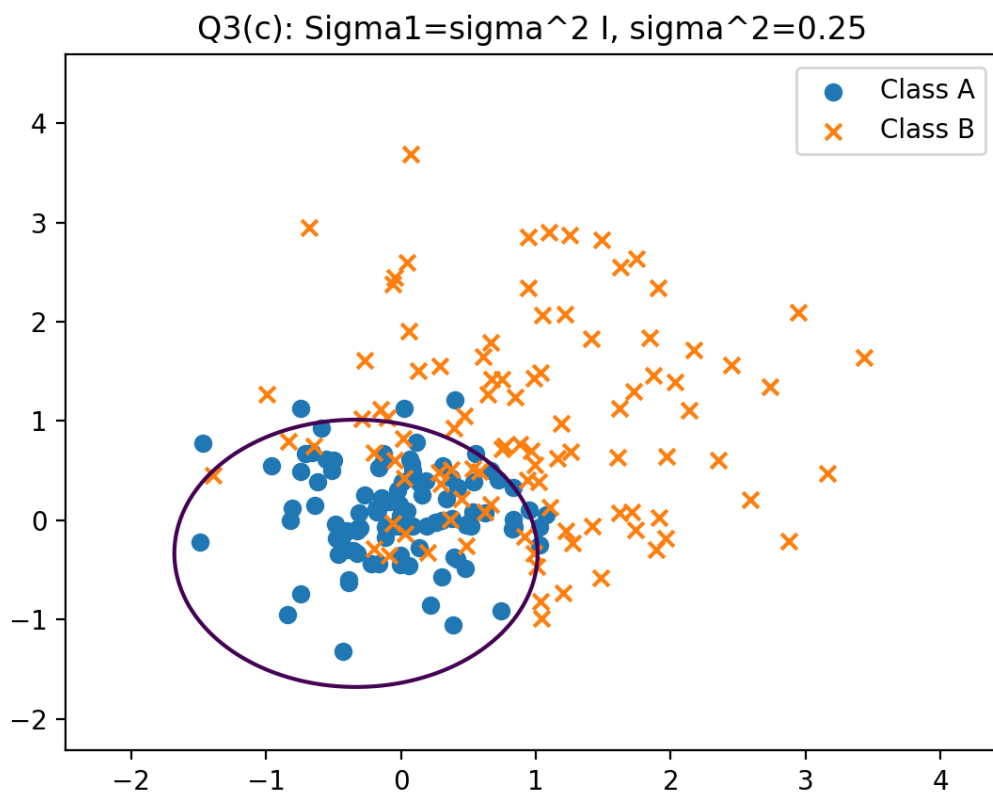


(c) optimal classifier:

$$\|x - \mu_2\|^2 - \frac{1}{\sigma^2} \|x - \mu_1\|^2 + 2 \log \frac{p}{1-p} - d \log(\sigma^2) > 0 \Rightarrow \text{decide } A,$$

(d) The shape is an ellipse.

(e) (a)(b) can't while (c) (d) can. Because GNB assumes conditional independence of features given the class, yielding a quadratic boundary in general, and a hypersphere when $\mu_1 = \mu_2$. (a) and (b) and degenerated cases where the boundary is linear.

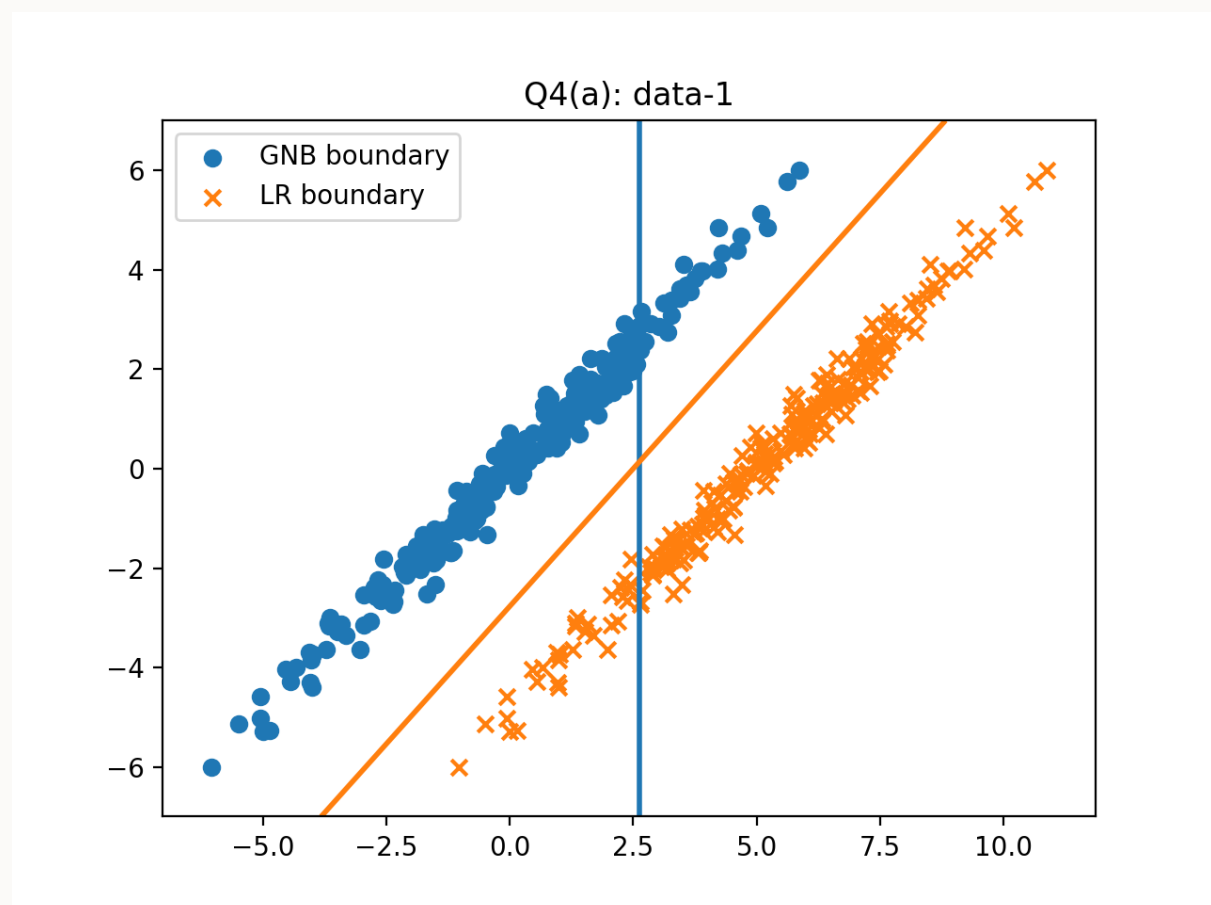


Assignment 4 (30 points): *Naive Bayes Classifier vs Linear Classifier* In this question, we have provided you with three carefully designed binary classification datasets (*data-1.npy*, *data-2.npy*, and *data-3.npy*).

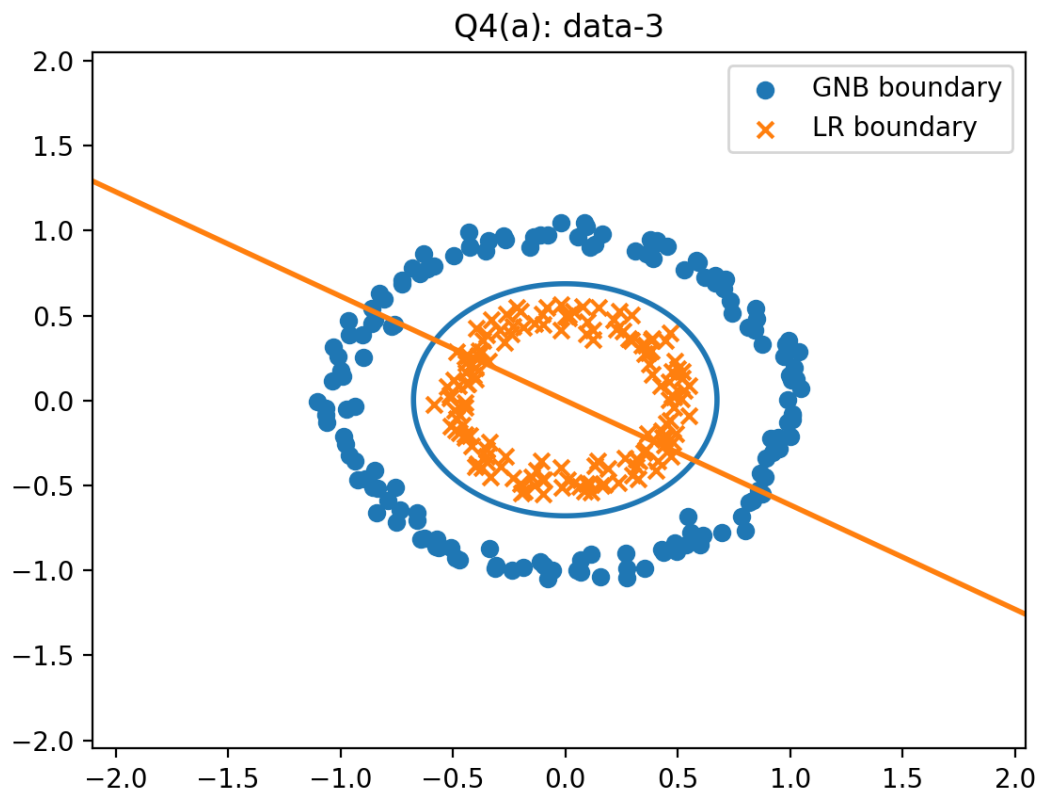
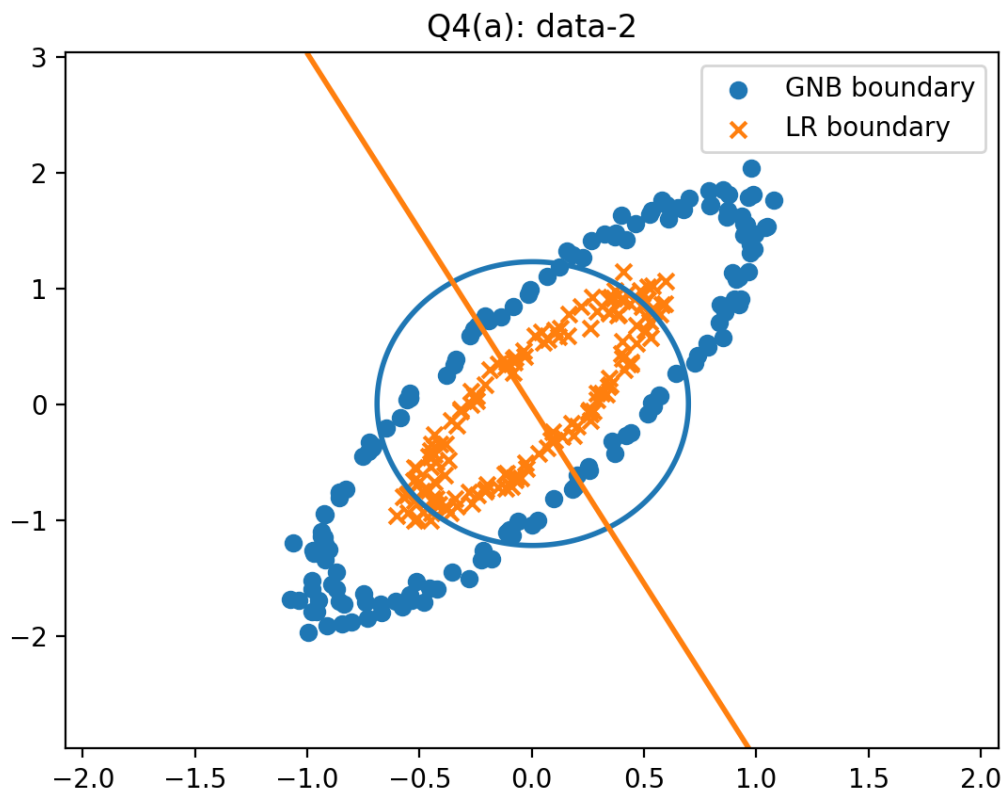
(a) (15 points) For each of the three datasets provided, fit a Gaussian Naive Bayes classifier and a linear logistic regression classifier using the *sklearn* library in Python. Plot the dataset with points color-coded according to their true labels. Additionally, plot the decision boundaries for both classifiers. (Hint: You may use *matplotlib* for plotting and *sklearn*'s *GaussianNB* and *LogisticRegression* for model fitting. The first two columns in the dataset represents the features and the last column is labels.)

(b) (15 points) Discuss why the Naive Bayes classifier and / or the linear classifier fail on each example. Justify your answers.

Solution. (a)



(b) data-1 (two lines; LR better): The class boundary is essentially linear. Logistic Regression fits a single straight separating hyperplane, which matches the geometry. GaussianNB assumes each class is Gaussian; if the data lie near thin lines, the Gaussian fit is a poor model, so its boundary can be biased.



data-2 (two ellipses; GNB better): The classes look like (possibly rotated) Gaussian clusters. GaussianNB models each class with a Gaussian and yields a curved/quadratic boundary, which fits elliptical structure. Logistic Regression is linear, so it cannot separate two ellipses well and underfits.

data-3 (two circles; GNB better): The class boundary is nonlinear (roughly circular). Logistic Regression's linear boundary fails. GaussianNB still works because each class is approximately Gaussian around its center, producing a quadratic boundary that can match circular separation.